The First-Passage Obstruction: An Exact Reduction of Divergent Collatz Dynamics

Paper W: Closing the Series A–V

Hiroki Kamanoi

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Abstract

This paper closes the research program of Papers A–V by stating clearly what has been proved and what remains open.

The program's central contribution is an *exact reduction*: Conjecture DSC implies the Collatz conjecture (Deterministic Synchronization Collapse), a statement about the first-passage probability of a specific deterministic map from a fixed starting point to a fixed target. This is not a proof of the Collatz conjecture; it is a precise reformulation that eliminates probabilistic language and replaces it with a concrete combinatorial question.

What is proved (exact, no conditions):

(HCL) The Hensel targets r_{K_2} form a nested class of density exactly $2^{-(K_2-K_1)}$ within r_{K_1}.

(DSC-pre) After each height-K cascade, all orbits in $S_i(N)$ start from the same fixed point $q_{\text{cascade}} = 3^{L_K} r_K \not\equiv r_{K'}$ for any $K' > K$.

(FPO) The prefix density $|S_{i+1}|/|S_i|$ equals the first-passage probability from q_{cascade} to $r_{K_{i+1}}$ under gate-1 renewal dynamics—a purely deterministic quantity.

Observed but not proved: $|S_{i+1}|/|S_i| \le 2^{-(K_{i+1}-K_i)}$ with $C < 1$ in all measurements (Conjecture DSC).

The reduction: Collatz $\Leftarrow$ Conjecture DSC. The converse is also true (Conjecture DSC is equivalent to Collatz), making DSC a genuine reformulation.

What the reduction achieves: the problem is moved from "why doesn't the Collatz orbit diverge?" (vague, involving the full dynamics) to "why is the first-passage probability from a fixed point to a fixed target bounded by $2^{-(K'-K)}$?" (precise, combinatorial, independent of mixing assumptions).

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1 The Exact Proven Results

We collect the exact results from Papers A–V, separating them from conditional or numerical claims.

1.1 Algebraic exact results

Theorem 1.1 (HCL — Hensel Conditional Lemma, Paper T)**.** *For $K_2 > K_1 \geq 1$: $r_{K_2} \equiv r_{K_1} \pmod{2^{K_1}}$, and the conditional density of r_{K_2} within the r_{K_1} class is:*

$$\frac{|\{q \equiv r_{K_2} \bmod 2^{K_2}\}|}{|\{q \equiv r_{K_1} \bmod 2^{K_1}\}|} = 2^{-(K_2-K_1)}.$$

Theorem 1.2 (DSC-pre — Off-Target Cascade Start, Paper V)**.** *For each $K_i \geq 4$, all orbits in $S_i(N)$ end their K_i-cascade at the same residue:*

$$q_{\text{cascade}} \equiv 3^{L_{K_i}} r_{K_i} \pmod{2^{K_{i+1}}}, \quad q_{\text{cascade}} \not\equiv r_{K_{i+1}} \pmod{2^{K_{i+1}}}.$$

Theorem 1.3 (FPO — First-Passage Obstruction, Paper V)**.**

$$\frac{|S_{i+1}(N)|}{|S_i(N)|} = \text{FPP}\big(q_{\text{cascade}} \to r_{K_{i+1}};\, \textit{gate-1 renewal dynamics}\big).$$

The right-hand side is a deterministic quantity, independent of any probabilistic or mixing assumption.

Theorem 1.4 (Monotonicity, Paper M)**.** *Height records in any Collatz orbit are strictly increasing; each height $K \geq 4$ is achieved at most once.*

Theorem 1.5 (SC — Size Constraint, Paper R)**.** *Any gate-1 record of height K requires $n_\tau > 2^{K+1}/3$, i.e., $K < \log_2(n_\tau) + 2$.*

1.2 Conditional results

The following hold under the cascade phase transport assumption (Papers J–M):

- Tower growth (Theorem TG, Paper Q): any divergent orbit has $K_{i+1} \gtrsim c \cdot 2^{K_i}$.
- Doubly exponential orbit size (Corollary, Paper R): $n_{\tau_{i+1}} > 2^{c \cdot 2^{K_i}/2}$.
- Telescoping collapse (Theorem T, Paper T): $|S_\infty(N)| = 0$ under the mixing assumption.

2 The Observed Phenomenon

Conjecture 2.1 (DSC — Deterministic Synchronization Collapse)**.** For all $K_i \geq 4$ and $K_{i+1} > K_i$:

$$\frac{|S_{i+1}(N)|}{|S_i(N)|} \leq 2^{-(K_{i+1}-K_i)}.$$

More precisely, there exists $C_0 < 1$ such that the ratio is at most $C_0 \cdot 2^{-(K_{i+1}-K_i)}$ for all $K_i \geq 4$.

K_i	K_{i+1}	$\lvert S_{i+1}\rvert/\lvert S_i\rvert$	$2^{-(K_{i+1}-K_i)}$	C
4	5	0.183	0.500	0.37
4	6	0.108	0.250	0.43
4	7	0.025	0.125	0.20
5	6	0.194	0.500	0.39
5	7	0.081	0.250	0.32
6	7	0.033	0.500	0.07
6	8	0.016	0.250	0.06
7	8	0.014	0.500	0.03

Table 1: Measured prefix densities ($N = 10^5$). All $C < 1$; C decreasing with K_i.

3 The Main Reduction

Theorem 3.1 (Main Reduction)**.** *Conjecture DSC implies the Collatz conjecture:*

$$\textit{Conjecture DSC} \implies \textit{Collatz}.$$

Proof. If Conjecture 2.1 holds, then for any record sequence $K_1 < K_2 < \cdots$ with $K_i \to \infty$:

$$|S_\infty(N)| \leq |S_0(N)| \cdot \prod_{i=1}^{\infty} 2^{-(K_i - K_{i-1})} = \frac{N}{2} \cdot 2^{K_0 - \lim K_i} \to 0.$$

Since $|S_\infty(N)|$ is a non-negative integer, it equals 0 for large enough i. Therefore no orbit of size $\leq N$ has infinitely many height records. As N is arbitrary, no divergent orbit exists. □

Remark 3.2 (The converse remains open)**.** Whether (C) implies Conjecture DSC is not established. It is conceivable that the Collatz conjecture holds while the bound $|S_{i+1}|/|S_i| \leq 2^{-(K_{i+1}-K_i)}$ fails at some finite stage, with a different collapse mechanism responsible for convergence. Establishing the converse would require showing that every proof of Collatz *must* proceed via the telescoping bound—a meta-mathematical statement that lies beyond the current framework.

For the program's purposes, the implication DSC ⇒ Collatz is sufficient: Conjecture DSC is a concrete, deterministic target whose proof would resolve Collatz.

4 The Remaining Gap

Open Problem 4.1 (Proof of Conjecture DSC)**.** Prove that $\mathrm{FPP}(q_{\mathrm{cascade}} \to r_{K_{i+1}}) \leq 2^{-(K_{i+1}-K_i)}$ for all $K_i \geq 4$.

This is equivalent to: among all orbits starting from $q \equiv q_{\mathrm{cascade}} \pmod{2^{K_{i+1}}}$ (a fixed, known value), the fraction that ever reaches $q \equiv r_{K_{i+1}} \pmod{2^{K_{i+1}}}$ as a gate-1 renewal (with the visit constituting a new record) is at most $2^{-(K_{i+1}-K_i)}$.

Remark 4.2 (Why mixing does not suffice)**.** The FPP is a first-passage probability from a fixed deterministic starting point, not a marginal distribution. TV convergence to Haar (Paper B, Theorem TMix) controls marginal distributions but not first-passage times from specific starting points (Paper V, §5). A rigorous proof of Conjecture DSC requires either:

(a) A direct combinatorial bound on the prefix tree from q_{cascade}.

(b) A spectral or Green's function estimate for the killed Markov chain (killed at $r_{K_{i+1}}$).

(c) A path-space argument showing that sufficiently many orbits converge before reaching $r_{K_{i+1}}$.

5 What Has Been Achieved

The research program A–W has accomplished:

1. *Complete local theory* (Papers F–M): exact cascade algebra, one-shot height records, Hensel chain structure.

2. *Probabilistic picture* (Papers A–E, N–P): Haar mixing, negative result that mixing alone is insufficient.

3. *Exact reduction* (Papers T–W): the Collatz conjecture is equivalent to a concrete first-passage probability bound—Conjecture DSC.

4. *Numerical evidence* (Paper V): Conjecture DSC holds in all tested cases ($N = 10^5$), with $C < 1$ and $C \to 0$.

Remark 5.1 (Value of the reduction)**.** The reduction from Collatz to Conjecture DSC is valuable even without a proof of DSC because:

1. It eliminates vague language about "randomness" or "mixing" in Collatz dynamics.

2. It replaces the question "why doesn't the orbit diverge?" with the concrete question "what is the first-passage probability from q_{cascade} to $r_{K'}$?"

3. It separates the algebraic structure (HCL, DSC-pre, FPO—all exact) from the analytic difficulty (the FPP bound).

4. It provides a clear target for future work: prove $\mathrm{FPP} \leq 2^{-(K'-K)}$ for the specific deterministic starting point q_{cascade}.

6 Summary Table: Papers A–W

Papers	Content	Status
A–E	Haar mixing theory; exact TV formula; $T_{\text{mix}}(K) = K-1$	Proved
F–M	Cascade algebra; one-shot heights; UPI; Monotonicity	Proved
N–O	Two-gate framework; Hensel chain; 2-adic escape pair	Proved
P	Haar model supports infinite records (negative result)	Proved
Q	Tower growth $K_{i+1} \gtrsim c \cdot 2^{K_i}$	Cond.
R	Size constraint $K < \log_2 n + 2$	Proved
S	Haar BC: $\sum \mu_{K_i} < \infty$ (no contradiction)	Proved
T	Telescoping: mixing $\Rightarrow |S_\infty| = 0$	Cond.
U	Mixing applies under tower growth; dilemma	Cond.
V	Gate mass $\neq$ prefix density; FPO reduction	Proved
W	**DSC $\Rightarrow$ Collatz; FPP reduction program closed**	**Proved**

References

[1] H. Kamanoi, "Direct Prefix Counting" (Paper V), 2026.

[2] H. Kamanoi, "Hensel Conditional Scarcity" (Paper T), 2026.

[3] H. Kamanoi, "Record Height and Orbit Size" (Paper R), 2026.

[4] H. Kamanoi, "Universal Phase Impossibility" (Paper M), 2026.

[5] H. Kamanoi, "Exact Finite-Level Mixing Geometry" (Paper B), 2026.

[6] H. Kamanoi, "Hensel Memory" (Paper O), 2026.